$$\sup\left\{\frac{\|T(x)\|}{\|x\|} \,\middle|\, x \in X\right\} < \infty$$

Define:

$$\|T\| = \sup\left\{\frac{\|T(x)\|}{\|x\|} \,\middle|\, x \in X,\ x \neq 0\right\}.$$

- $T \in B(X, Y)$

  $\frac{\|T(x)\|}{\|x\|} \leq \|T\|$, $\forall x \in X$

  $\Longrightarrow \underline{\|T(x)\| \leq \|T\|\,\|x\|}$.

$$\inf\left\{M \,\middle|\, \frac{\|T(x)\|}{\|x\|} \leq M,\ \forall x \neq 0 \in X\right\} = \|T\|$$

- Let $T_1 \in B(X, Y)$

  $T_2 \in B(X, Y)$

  $\|T_1(x)\| \leq \|T_1\|\,\|x\|$ $\quad \forall x \in X$

  $\|T_2(x)\| \leq \|T_2\|\,\|x\|$

consider

$$\begin{aligned}\|(T_1 + T_2)(x)\|_Y &= \|T_1(x) + T_2(x)\|_Y \\ &\leq \|T_1(x)\| + \|T_2(x)\| \\ &\leq \|T_1\|\,\|x\| + \|T_2\|\,\|x\| \\ &= (\|T_1\| + \|T_2\|)(\|x\|)\end{aligned}$$

we get

$$\frac{\|(T_1 + T_2)(x)\|}{\|x\|} \leq \|T_1\| + \|T_2\|$$

Now take supremum.

$$\Longrightarrow \|T_1 + T_2\| \leq \|T_1\| + \|T_2\|$$

where $T_1 + T_2 \in B(X, Y)$.

→ "$\|T\|$ is called operator norm"

THEOREM

$B(X,Y)$ endowed with operator norm is banach whenever Y is Banach.

Hence $B(X)$: bounded linear f^n from $X \to X$ is always banach if X is banach.

To prove that $B(X,Y)$ is complete,
Let (T_n) be a cauchy sequence in $B(X,Y)$
Given $\varepsilon_1 > 0$, $\exists N_{\varepsilon_1} \in \mathbb{N}$
s.t

$$\left.\begin{aligned} &\|T_n - T_m\| < \varepsilon_1 \\ &\quad \forall n,m \geq N_{\varepsilon_1} \end{aligned}\right] \to (1)$$

$\therefore$ For $x \in X$

$$\|T_n(x) - T_m(x)\| \leq \|T_n - T_m\| \, \|x\|$$

$$\leq \varepsilon_1 \|x\| \qquad -(2)$$

$$\forall n,m > N_{\varepsilon_1}$$

Given $\varepsilon > 0$

let $\varepsilon_1 = \dfrac{\varepsilon}{\|x\|}$

From (2),

$$\|T_n(x) - T_m(x)\| < \varepsilon \quad , \forall n,m \geq N_{\varepsilon_1}$$

we get $(T_n(x))$ is a cauchy seq^n in Y.
Y is banach (given)
$(T_n(x))$ converges as $n \to \infty$

Define $T: X \to Y$

$$T(x) = \lim_{n\to\infty} T_n(x) \quad , \forall x \in X$$

let $m \to \infty$

$\|T_n(x) - T(x)\| \leq \varepsilon_1 \|x\|$

$\forall\, n \geq N_{\varepsilon_1}$

$\Rightarrow \sup_{\substack{x \in X \\ x \neq 0}} \dfrac{\|T_n(x) - T(x)\|}{\|x\|} \leq \varepsilon_1$

$\forall\, n \geq N_{\varepsilon_1}$

$\Rightarrow \|T_n - T\| \leq \varepsilon_1$, $\forall\, n \geq N_{\varepsilon_1}$

$\therefore \|T_n - T\| \to 0$ as $n \to \infty$

& $T \in B(X, Y)$.

$\therefore$ $B(X, Y)$ is a Banach space.

BOUNDED LINEAR MAPS

- $T \in B(X, Y)$, $\exists\, M > 0$ s.t $\|T(x)\| \leq M\|x\|$.

Operator Norm

$$\|T\|_{op} = \sup_{\substack{x \in X \\ x \neq 0}} \frac{\|T(x)\|_Y}{\|x\|_X}$$

- $(B(X, Y), \|\cdot\|_{op})$ is a NLS.

- If Y is Banach space, then $B(X, Y)$ is Banach.

- $B(X) =$ Algebra
  $\| B(X, X)$
  so we can say it is a banach algebra, if X is complete.
- $B(X, \mathbb{K})$ is always a Banach space.
  $X' = B(X, \mathbb{K}) =$ dual space of X.
  $f : X \to \mathbb{K}$ such that ① f is linear
  ② $|f(x)| \leq M\|x\|$

  ③ $\|f\| = \sup_{\substack{x \in X \\ x \neq 0}} \dfrac{|f(x)|}{\|x\|}$.

Exercise $T \in B(X, Y)$

$$\|T\| = \sup_{\substack{x \in X \\ x \neq 0}} \frac{\|T(x)\|}{\|x\|} \stackrel{?}{=} \sup_{\|z\| < 1} \|T(z)\| \stackrel{?}{=} \sup_{\|z\| = 1} \|T(z)\| \quad X \neq 0$$

→ All 3 are equivalent.

- $T \in B(X, Y)$
  $\|T\| \leq M$
  If $\exists\, x_0 \in X$ s.t
  $\|T(x_0)\| = M\|x_0\|$
  then $\boxed{\|T\| = M}$

- Suppose X, Y are two sets
  & $T : X \longrightarrow Y$.
  $X_0 \subseteq X$
  Restriction of T to X_0 is denoted by $T|_{X_0} : X_0 \to Y$.

  where $T|_{X_0}(x) = T(x)$, $\forall x \in X_0$.

- X, Y are two sets, $X \subseteq M$
  Let $f : X \longrightarrow Y$ be a function.
  An extention of f to M is a function
  $\tilde{f} : M \longrightarrow Y$ s.t
  $\tilde{f}(x) = f(x)$, $\forall x \in X$

  $\tilde{f}|_X = f$.

- T is a linear transformation.

$$T(x+y) = \lim_{n\to\infty} T_n(x+y) = \lim_{n\to\infty} (T_n(x) + T_n(y))$$

$$= \lim_{n\to\infty} T_n(x) + \lim_{n\to\infty} T_n(y)$$

$$= T(x) + T(y)$$

$T(\alpha x) = \alpha T(x)$ (check)

$\therefore T \in L(X, Y)$

Note that (T_n) is a C.S in $B(X,Y)$

$$\big| \|T_n\| - \|T_m\| \big| \leq \|T_n - T_m\|$$
$$\leq \varepsilon_1, \forall n, m \geq N_{\varepsilon_1}$$

Thus $(\|T_n\|)$ is a cauchy seq^n of real numbers i.e $\exists M > 0$ s.t $\|T_n\| \leq M \; \forall n \in \mathbb{N}$ — ③

For $x \in X$

$$\|T_n(x)\| \leq \|T_n\| \, \|x\|$$
$$\leq M \|x\| \text{ (by ③)}.$$

$$\lim_{n\to\infty} \|T_n(x)\| \leq M\|x\|$$

$$\Big\| \lim_{n\to\infty} T_n(x) \Big\| \leq M \|x\|$$

$\Rightarrow \|T(x)\| \leq M\|x\|$

$\Rightarrow T$ is bounded

$\therefore T \in B(X,Y)$

<u>CLAIM</u>: $T_n \to T$ as $n \to \infty$ in operator norm

$\|T_n - T\| \to 0$ as $n \to \infty$.

Let $x \in X$, $x \neq 0$

$$\|T_n(x) - T_m(x)\| = \|(T_n - T_m)(x)\| \leq \|T_n - T_m\| \, \|x\|$$
$$\leq \varepsilon_1 \|x\|, \forall n, m \geq N_{\varepsilon_1}$$

__/__/__

If even & odd terms of a seq^n converge to same limit then seq^n converges to that limit.

$z_m \to x$ as $m \to \infty$.

$\therefore z_m$ is cauchy seq^n $\Rightarrow T(z_m)$ is cauchy. and $T(z_m)$ is convergent

Again $T(x_n)$ & $T(y_n)$ converge to same limit so $T(z_m)$ also converge to that

$\therefore \tilde{T}$ is well defined.

- $\tilde{T}$ is a linear map. (Exercise).

- $\tilde{T}$ is bounded.

$\|\tilde{T}(x)\| = \|y\| = \|\lim_{n\to\infty} T(x_n)\|$ where $x_n \to x$ so $\|x_n\| \to \|x\|$.

$= \lim_{n\to\infty} \|T(x_n)\| \leq \lim_{n\to\infty} \|T\| \|x_n\|$

$= \|T\| \|x\| \quad \forall x \in X$

so $\tilde{T} \in B(X, Y)$.

- $\|\tilde{T}\| = \|T\|$.

From the previous proof

$\|\tilde{T}(x)\| \leq \|T\| \|x\|$

$\dfrac{\|\tilde{T}(x)\|}{\|x\|} \leq \|T\|$

Now take supremum,

$\rightarrow \|\tilde{T}\| \leq \|T\|$

THEOREM. X, Y be NLS
Assume $X \subseteq M$ where M is a NLS
and Y is a Banach space. Suppose $T: X \to Y$ is a bounded linear map
Then $\exists$ a bounded linear map

$\tilde{T}: \overline{X} \to Y$ s.t
$\tilde{T}(x) = T(x)$, $\forall x \in X$

$\|\tilde{T}\| = \|T\|$.

Proof: Pick $x \in \overline{X}$
$\exists (x_n) \in X$ s.t $x_n \to x$ as $n \to \infty$
$\therefore (x_n)$ is a cauchy seq^n
$\|T(x_m) - T(x_n)\|$
$= \|T(x_m - x_n)\| \leq \|T\| \, \|x_n - x_m\|$
Thus $\{T(x_n)\}$ is a cauchy sequence in Y.
$\therefore T(x_n)$ converges in Y.
Let $\lim_{n \to \infty} T(x_n) = y$, $y \in Y$.

Define $\tilde{T}: \overline{X} \to Y$
$\tilde{T} x = y$

- $\tilde{T}$ is well defined.

Suppose $\exists (y_n)$ s.t $y_n \to x$ as $n \to \infty$
we have $x_n \to x$ as $n \to \infty$

consider

$z_{2k} = y_k$
$z_{2k-1} = x_k$
$(z_m) = (x_1, y_1, x_2, y_2, x_3, y_3 \ldots)$

$\|x_n - y_n\| \to \boxed{0}$

CLAIM: $\|\tilde{T}\| \geq \|T\|$

$$\|\tilde{T}\| = \sup\left\{\frac{\|\tilde{T}(x)\|}{\|x\|}, \begin{array}{l} x \in \overline{X} \\ x \neq 0 \end{array}\right\}.$$

$$\geq \sup\left\{\frac{\|\tilde{T}(x)\|}{\|x\|}, \begin{array}{l} x \in X \\ x \neq 0 \end{array}\right\}$$

$$= \sup\left\{\frac{\|T(x)\|}{\|x\|}, \begin{array}{l} x \in X \\ x \neq 0 \end{array}\right\}.$$

$$= \|T\|.$$

- Extention is ~~e~~ unique.

$\tilde{T} : \overline{X} \to Y$

$\hat{T} : \overline{X} \to Y$

$\tilde{T}(x) = \lim_{n\to\infty} T(x_n)$ where $x_n \to x$ as $n \to \infty$.

$= \lim_{n\to\infty} \hat{T}(x_n)$. because $\hat{T}(x_n) = \tilde{T}(x_n) = T(x_n)$ since $x_n \in X$.

$= \hat{T}\left(\lim_{n\to\infty} x_n\right)$.

$= \hat{T}(x)$. $\forall x \in X$.